



CANDIDATE
NAME

--

CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--

0620/63

October/November 2021

1 hour

No additional materials are needed.

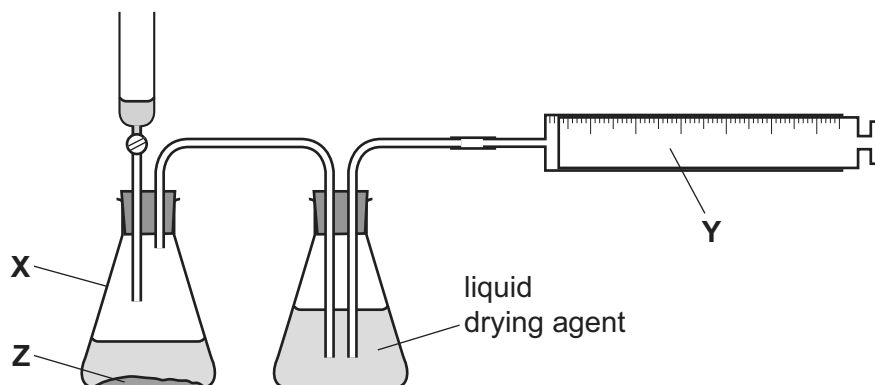
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages. Any blank pages are indicated.

- 1 Hot concentrated hydrochloric acid reacts with solid manganese(IV) oxide to make chlorine gas. Chlorine gas can be dried by bubbling it through a liquid drying agent.

The diagram shows the apparatus used to make and collect a sample of dry chlorine gas. There is one error in the diagram.



- (a) Name the items of apparatus labelled X and Y.

X
 Y [2]

- (b) Name the substance labelled Z.

..... [1]

- (c) On the diagram draw one arrow to show where heat should be applied so that chlorine gas is made. [1]

- (d) There is one error in the way the apparatus has been set up.

- (i) On the diagram draw a circle around the error in the apparatus. [1]

- (ii) Describe what would happen if the apparatus is used before the error is corrected.

.....
 [1]

[Total: 6]

- 2 A student investigated the temperature change when zinc reacted with two different aqueous solutions of copper(II) sulfate, solution **Q** and solution **R**.

Two experiments were done.

(a) Experiment 1

- A polystyrene cup was placed in a 250 cm³ beaker for support.
- Using a measuring cylinder, 25 cm³ of solution **Q** was poured into the polystyrene cup.
- Using a thermometer, the initial temperature of solution **Q** was measured.
- 3 g of zinc powder was added to the polystyrene cup. At the same time a stop-clock was started.
- Using the thermometer, the mixture in the polystyrene cup was continually stirred and the temperature measured every 30 seconds.

initial temperature in Experiment 1	23 °C
-------------------------------------	-------

Use the thermometer diagrams and the initial temperature to complete the table.
Calculate the temperature changes using the equation:

$$\text{temperature change} = \text{temperature} - \text{initial temperature}$$

time/s	30	60	90	120	150	180	210	240
thermometer diagram								
temperature/°C								
temperature change/°C								

[3]

(b) Experiment 2

- The polystyrene cup was washed out with distilled water.
- Experiment 1 was repeated using solution **R** instead of solution **Q**.

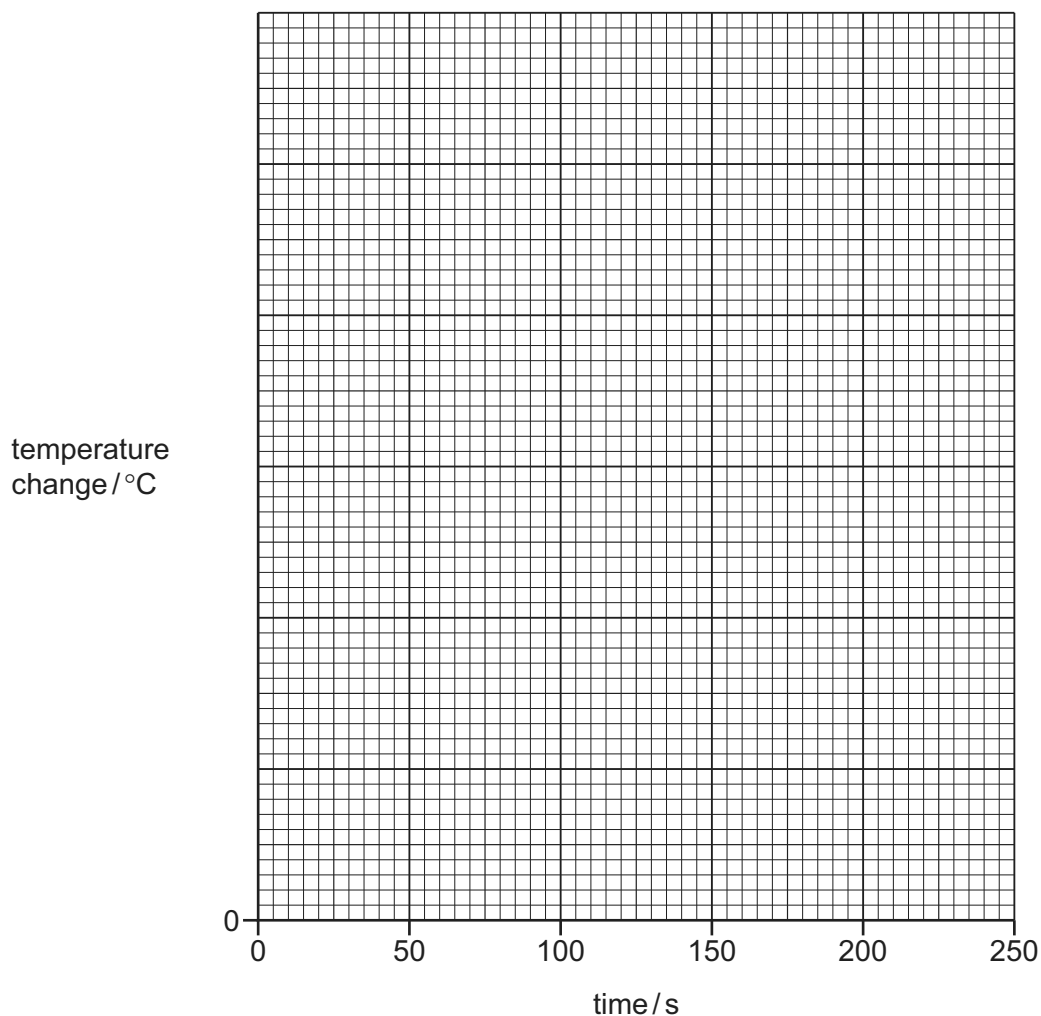
initial temperature in Experiment 2	24 °C
-------------------------------------	-------

Use the thermometer diagrams and the initial temperature to complete the table.

time / s	30	60	90	120	150	180	210	240
thermometer diagram								
temperature / °C								
temperature change / °C								

[3]

- (c)** Complete a suitable scale on the y-axis and plot the results from Experiment 1 and Experiment 2 on the grid. Draw two curves of best fit. Both curves must start at (0,0). Label your curves.



[5]

- (d) From your graph, deduce the temperature change at 110 seconds in Experiment 1.

Show clearly **on the grid** how you worked out your answer.

..... °C
[2]

- (e) Predict the temperature of the solution in Experiment 2 after 5 hours. Explain your answer.

.....
..... [2]

- (f) (i) Suggest why the experiments were done in a polystyrene cup rather than a glass beaker.

..... [1]

- (ii) Describe how the results would be different if a glass beaker is used in place of the polystyrene cup.

..... [1]

- (g) Suggest **one** change that could be made to the apparatus that would improve the accuracy of the results. Explain why this change would improve the accuracy of the results.

change

explanation

.....
[2]

[Total: 19]

- 3 Solid **S** and solid **T** were analysed.
Tests were done on each substance.

tests on solid S

tests	observations
test 1 Solid S was placed in a boiling tube and 10 cm ³ of dilute hydrochloric acid was added.	effervescence
The solution formed in test 1 was decanted from the remaining solid S . The solution is solution U . test 2 Aqueous sodium hydroxide was added dropwise and then in excess to solution U .	white precipitate, insoluble in excess

The gas given off in **test 1** was carbon dioxide.

- (a) Describe how the gas produced in **test 1** could be tested to show that it was carbon dioxide. Give the expected result of the test.

test

result

[2]

- (b) Identify solid **S**.

.....

..... [2]

tests on solid T

Solid **T** was iron(III) chloride.

Solid **T** was dissolved in water to form solution **T**. Solution **T** was divided into four equal portions in four test-tubes.

- (c) To the first portion of solution **T**, aqueous sodium hydroxide was added dropwise and then in excess.

observations
..... [2]

- (d) To the second portion of solution **T**, 2 cm³ of aqueous ammonia was added.

observations [1]

- (e) To the third portion of solution **T**, 1 cm³ of dilute nitric acid followed by a few drops of aqueous silver nitrate were added.

observations [1]

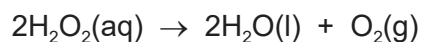
- (f) To the fourth portion of solution **T**, 1 cm³ of dilute nitric acid followed by a few drops of aqueous barium nitrate were added.

observations [1]

[Total: 9]

- 4 Catalysts are substances which increase the rate of a reaction but are unchanged at the end of the reaction.

Aqueous hydrogen peroxide decomposes slowly to form water and oxygen.



Copper(II) oxide is an insoluble solid.

Plan an investigation to find out if copper(II) oxide is a catalyst for the decomposition of hydrogen peroxide. You must include how your results will tell you if copper(II) oxide is a catalyst. You have access to copper(II) oxide, aqueous hydrogen peroxide and all normal laboratory apparatus.

[6]

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.